

koloHart: Neck-Worn Tactile Choker for Sensory Substitution and Sensory Enhancement

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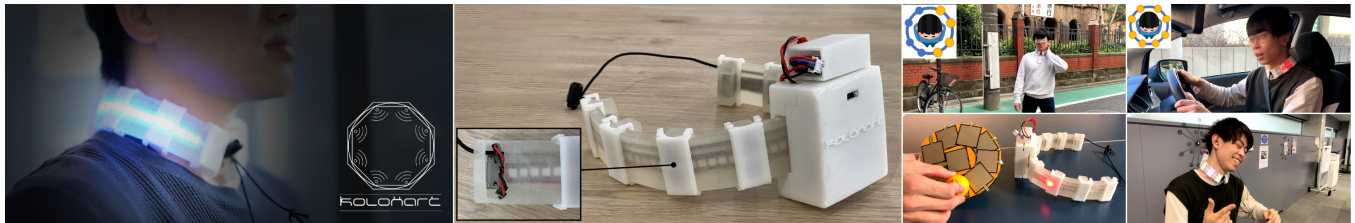


Figure 1: koloHart, a novel neck-worn tactile choker designed to leverage the latent potential of the neck for sensory substitution and sensory enhancement.

Abstract

Tactile feedback on the neck is considered to hold great potential due to its unique sensory characteristics as a sensory organ. However, this area remains largely unexplored at present. To address this potential, we propose koloHart, a neck-worn tactile choker designed to utilize the neck's distinctive properties for both sensory substitution and sensory enhancement. koloHart is a wearable haptics device equipped with eight linear resonant actuators (LRAs) arranged around the neck, capable of delivering fine vibrotactile feedback from 360-degree directions. This poster showcases a variety of potential application scenarios that highlight the expressive and immersive potential of neck-based tactile feedback.

CCS Concepts

• **Hardware** → **Emerging technologies**; • **Human-centered computing** → **Haptic devices**.

Keywords

Tactile, Wearable Haptics, Vibrotactile, Neck-Worn Device, Sensory Substitution, Sensory Enhancement

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1 Introduction

In recent years, tactile devices have been utilized across a wide range of applications, gaining attention as tools for enhancing usability and enabling user experiences. While the development of such devices has advanced significantly, one area of the body that remains relatively unexplored for tactile feedback is the neck.

The neck possesses unique characteristics as a sensory organ. It keeps both hands free, has high vibration sensitivity, and can perceive stimuli from all directions. Its dense afferent nerves enable rapid reflexive responses, while deep proprioception allows users to sense vibrations within the body's core. Psychologically, the neck is closely tied to emotional expression and exhibits qualities not found in other body parts. These features suggest that the neck holds significant potential for tactile interaction. While there have been some initial explorations [Matsuda et al. 2020; Schaack et al. 2019], it still remains a largely unexplored and untapped domain.

In this work, we propose koloHart, a neck-worn tactile choker designed to leverage the latent potential of the neck for both sensory substitution and sensory enhancement. koloHart is a wearable haptics device equipped with eight linear resonant actuators (LRAs) arranged around the neck, capable of delivering fine vibrotactile feedback from 360-degree directions. For sensory substitution, koloHart provides directional navigation cues and hazard alerts by supplementing visual or auditory information through tactile feedback on the neck, enabling safer and smoother everyday actions without relying on conventional senses. For sensory enhancement, it supports applications such as sports skill improvement and immersive experiences through music or VR by presenting non-visual, non-auditory information via the skin. These interactions aim to enhance immersion, facilitate the acquisition of new bodily skills, and expand human perception.

This poster showcases potential application scenarios that leverage the unique characteristics of neck-based tactile feedback.

2 koloHart

2.1 System Design

Fig. 2 shows an overview and the system design of koloHart. Eight vibration actuators are arranged around the neck, with both the processing unit and power supply fully integrated into the device. This configuration enables koloHart to operate as a self-contained system. In addition, the system includes I/O and communication interfaces, enabling a highly customizable architecture adaptable to a wide range of applications.

koloHart adopts the HAPTIC REACTOR (Hybrid Tough Type) developed by Alps Alpine as its vibration actuator. This linear resonant actuator (LRA) offers significantly lower latency and higher responsiveness compared to conventional eccentric rotating mass (ERM) vibration motors. With dual resonance frequencies at 160 Hz and 320 Hz, the actuator can produce bidirectional vibration and supports a wide frequency range for rich vibrotactile expression.

The hardware exterior includes modular LRAs embedded in a molded silicone belt. The use of silicone suppresses vibration propagation between actuators, allowing each unit to deliver localized, independent feedback. The device is worn over the head and secured with an elastic cord, allowing any user to put it on in a few seconds. It naturally settles at the base of the neck, where the weight is distributed across the shoulders, enhancing comfort.

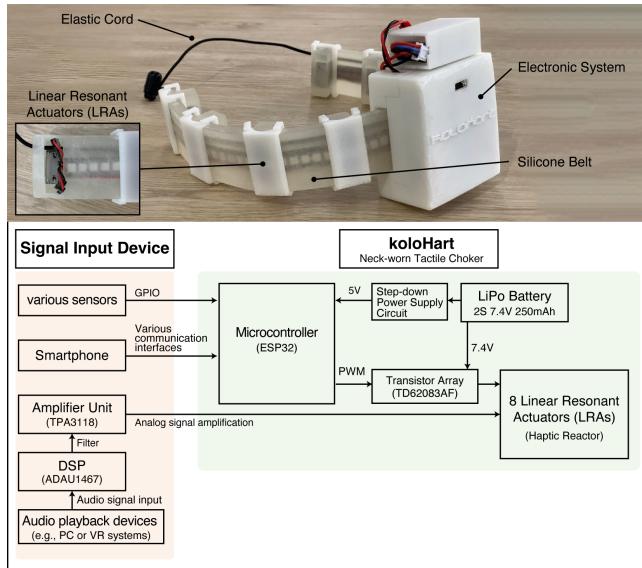


Figure 2: (Top) Overall view. (Bottom) System design.

2.2 Directional Tactile Guidance Using Pseudo-Force and Phantom Sensation

We combine two haptic techniques, pseudo-force sensation [Rekimoto 2013] and phantom sensation [von Békésy 1959], to achieve directional tactile feedback on the neck. Pseudo-force sensation is a technique that induces the illusion of force in a specific direction by introducing asymmetry in the intensity or timing of vibrations. This is applied to a neck-worn tactile choker to provide directional

cues that guide the user's attention or movement toward a specific direction. On the other hand, phantom sensation is a perceptual phenomenon in which a user feels vibration at a location where no actuator exists, typically between two stimulated points on the skin. This allows us to create the illusion of continuous motion or presence of a virtual actuator. To smoothly interpolate tactile cues in different directions, we define a linear mapping between the intended direction of vibration and the PWM duty ratios applied to each actuator. Fig. 3 presents (Left) the linear mapping between the target direction of vibration and the PWM duty ratio applied to each actuator, and (Right) an overview of the neck-based tactile feedback mechanism used for directional guidance.

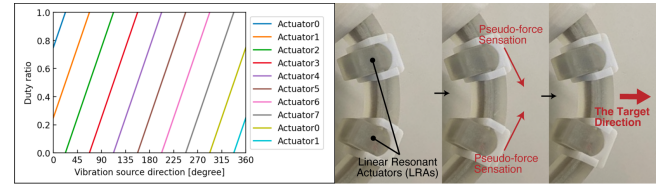


Figure 3: (Left) The linear mapping between the target direction of vibration and the PWM duty ratio applied to each actuator. (Right) Overview of the neck-based tactile feedback mechanism used for directional guidance.

2.3 Spatial and Immersive Tactile Experience Driven by Audio Signals

koloHart uses 2- and 8-channel audio sources to deliver spatial vibrotactile feedback around the neck, creating an immersive experience. Users feel as if enveloped in sound and may perceive illusions such as presence from behind. To achieve this, the system incorporates an audio playback device (e.g., PC or VR), a digital signal processor (DSP), and an amplifier unit. The embedded linear resonant actuators (LRAs) have resonance peaks at 160 Hz and 320 Hz, supporting rich tactile expression across a broad frequency range. However, when input signals exceed these resonance frequencies, the actuators cease to vibrate effectively, and even low-frequency performance may be degraded. To address this, we implement a low-pass filter on the DSP to attenuate frequency components above 320 Hz. When high-frequency information needs to be conveyed through tactile feedback, the signal is pitch-shifted down to a lower frequency band before being sent to each LRA. Through this combination of pitch conversion, signal summation, and other high-speed, flexible DSP-based processing techniques, the system enables responsive tactile presentation compatible with a wide variety of audio sources.

3 Conclusion and Future Work

In this study, we proposed koloHart, a neck-worn tactile choker designed for sensory substitution and sensory enhancement, focusing on the neck as a body part with untapped potential. In future work, we will continue to investigate novel use cases of neck-based haptics by examining its effectiveness in application areas such as medical care, rehabilitation, navigation, and remote communication.

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